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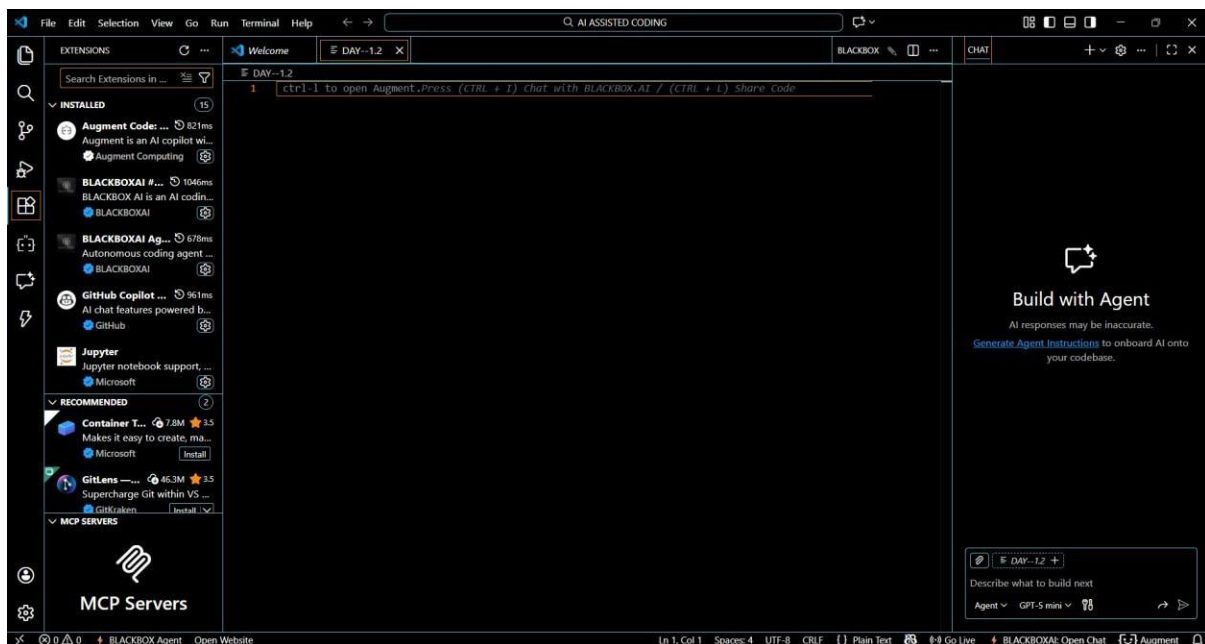
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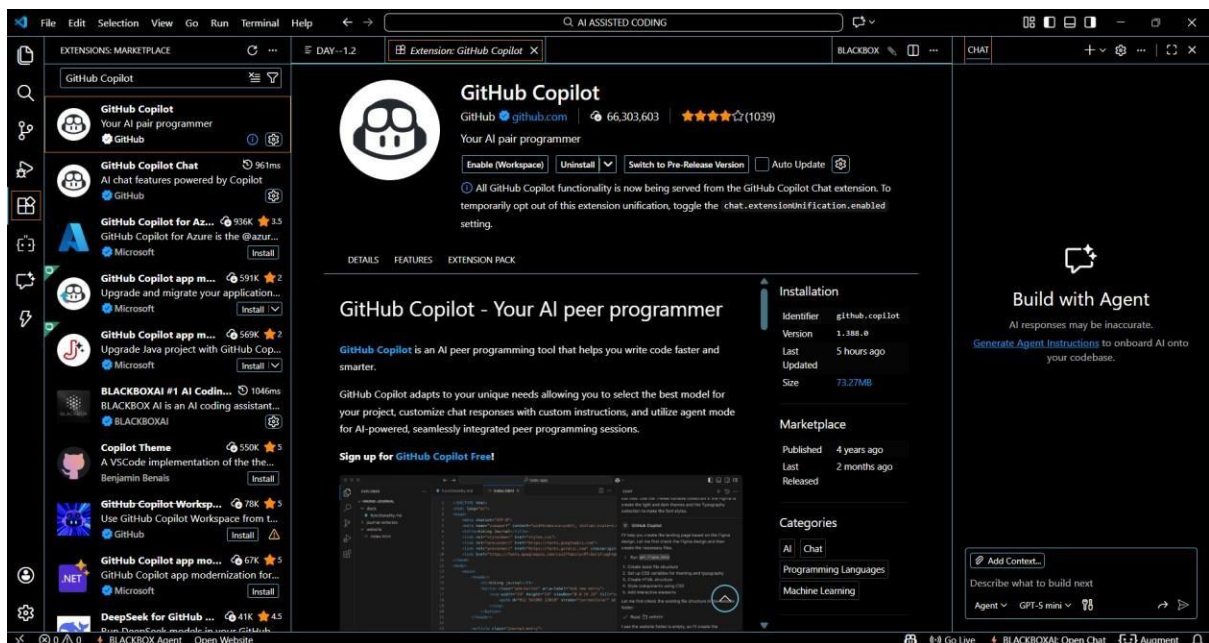
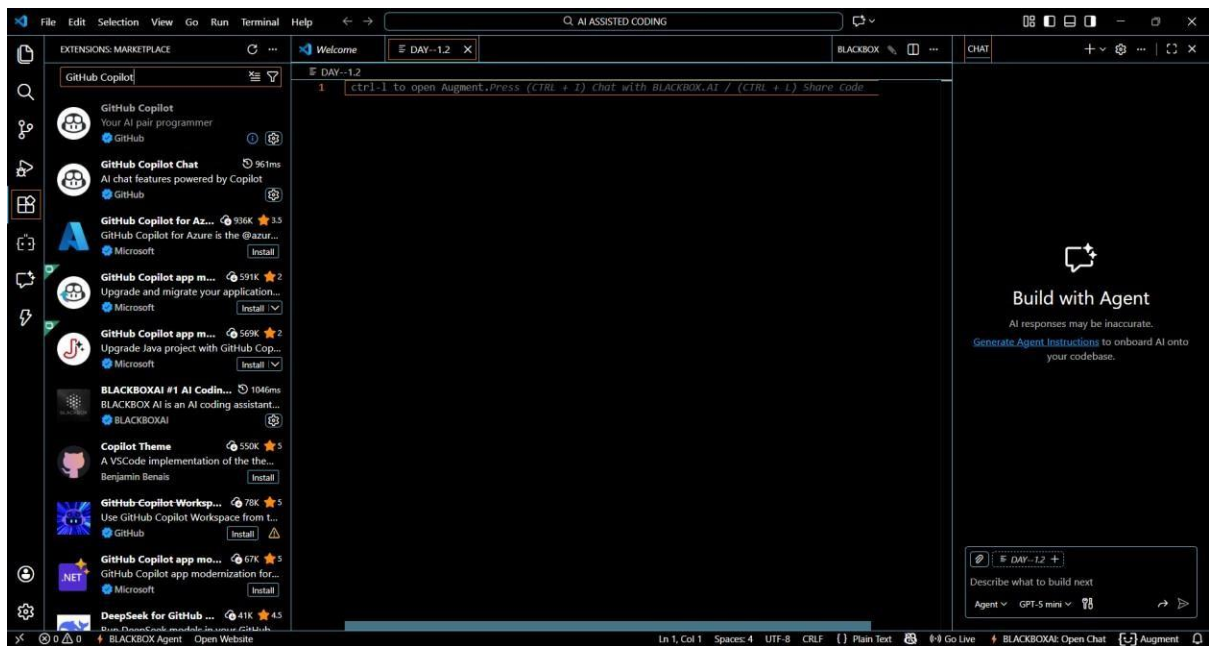
Lab 1.5: AI-Assisted Coding using GitHub Copilot

Task 0: Environment Setup

Steps:

1. Install Visual Studio Code
2. Open VS Code → Extensions
3. Search GitHub Copilot
4. Click Install
5. Sign in with GitHub account
6. Enable Copilot suggestions





Explanation: GitHub Copilot was installed and configured in Visual Studio Code by signing in with a GitHub account. This enables AI-based code suggestions directly inside the editor, helping developers write code faster and more efficiently.

Task 1: String Reversal Without Functions

Prompt:

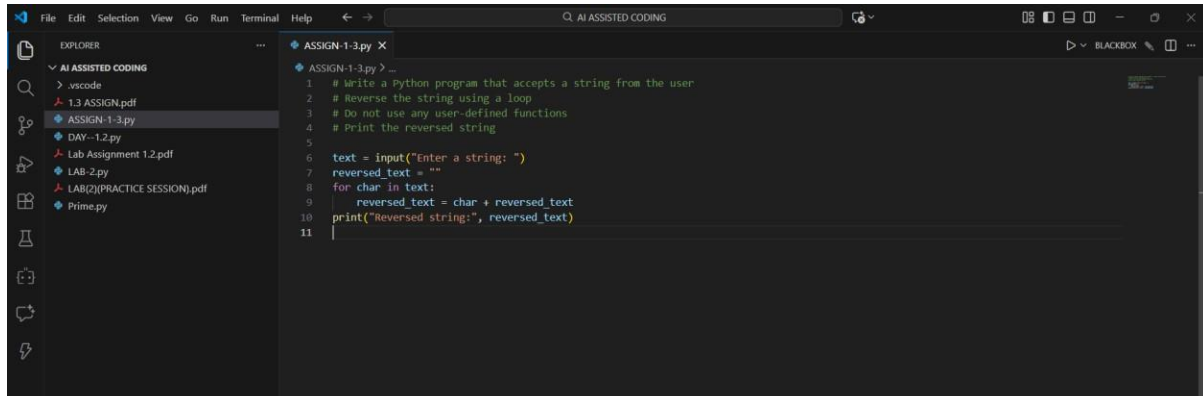
Write a Python program that accepts a string from the user

Reverse the string using a loop

Do not use any user-defined functions

Print the reversed string

###CODE:

A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a file named 'ASSIGN-1-3.py' selected. The main editor window displays the code for 'ASSIGN-1-3.py'. The code includes comments and a loop-based implementation for reversing a string.

```
1 # Write a Python program that accepts a string from the user
2 # Reverse the string using a loop
3 # Do not use any user-defined functions
4 # Print the reversed string
5
6 text = input("Enter a string: ")
7 reversed_text = ""
8 for char in text:
9     reversed_text = char + reversed_text
10 print("Reversed string:", reversed_text)
11
```

OUTPUT:

A screenshot of a terminal window. The terminal shows the execution of the Python script. The user enters '4' as the string, and the program outputs 'Reversed string: 4'.

```
PS C:\Users\pc\OneDrive\Desktop\Devops> & C:/Users/pc/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/pc/Downloads/Untitled-1.py
Enter a string: 4
Reversed string: 4
PS C:\Users\pc\OneDrive\Desktop\Devops>
```

Explanation: In this task, GitHub Copilot generated Python code to reverse a string using a loop without defining any functions. The logic was written directly in the main program, demonstrating basic procedural programming.

Task 2: Code Optimization s Readability

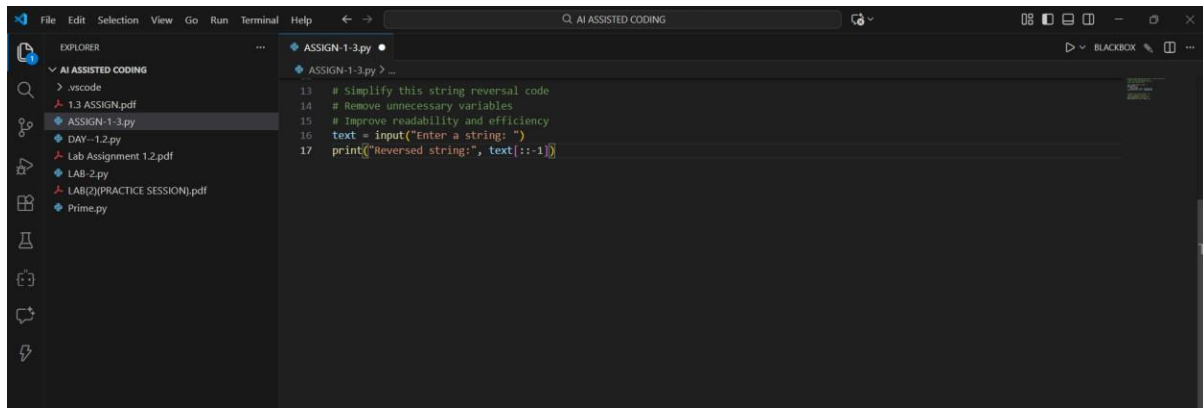
Prompt:

Simplify this string reversal code

Remove unnecessary variables

Improve readability and efficiency

###CODE:



```
13 # Simplify this string reversal code
14 # Remove unnecessary variables
15 # Improve readability and efficiency
16 text = input("Enter a string: ")
17 print("Reversed string:", text[::-1])
```

OUTPUT:



```
PS C:\Users\pc\OneDrive\Desktop\Devops> & C:/Users/pc/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/pc/Downloads/Untitled-1.py
Enter a string: 2
Reversed string: 2
PS C:\Users\pc\OneDrive\Desktop\Devops>
```

Explanation: The Copilot-generated code was optimized by simplifying the logic and removing unnecessary variables. The improved version produces the same output with better readability and reduced code complexity.

Task 3: String Reversal Using Functions

Prompt:

- # Write a Python program using a function to reverse a string
- # The function should return the reversed string
- # Add meaningful comments

###CODE:

```

18
19 # Write a Python program using a function to reverse a string
20 # The function should return the reversed string
21 # Add meaningful comments
22 def reverse_string(text):
23     # This function reverses the given string
24     return text[::-1]
25
26 user_input = input("Enter a string: ")
27 result = reverse_string(user_input)
28 print("Reversed string:", result)

```

OUTPUT:

```

PS C:\Users\pc\OneDrive\Desktop\Devops> & C:\Users\pc\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:\Users\pc\Downloads\Untitled-1.py
Enter a string: 6
Reversed string: 6
PS C:\Users\pc\OneDrive\Desktop\Devops>

```

Explanation: GitHub Copilot was used to create a modular program using a user-defined function to reverse a string. This approach improves reusability, clarity, and makes the code easier to maintain.

Task 4: Procedural vs Modular Comparison

Step 1: Prompt:

Compare string reversal programs with and without functions

Discuss clarity, reusability, debugging, and scalability

Step 2: Answer (Analysis Output)

Feature	Without Function	With Function
Code clarity	Medium	High
Reusability	Low	High
Debugging	Difficult	Easy
Large applications	Not suitable	Suitable

Explanation: A comparison was made between function-based and non-function-based programs. The analysis shows that modular code is more reusable, easier to debug, and better suited for large-scale applications.

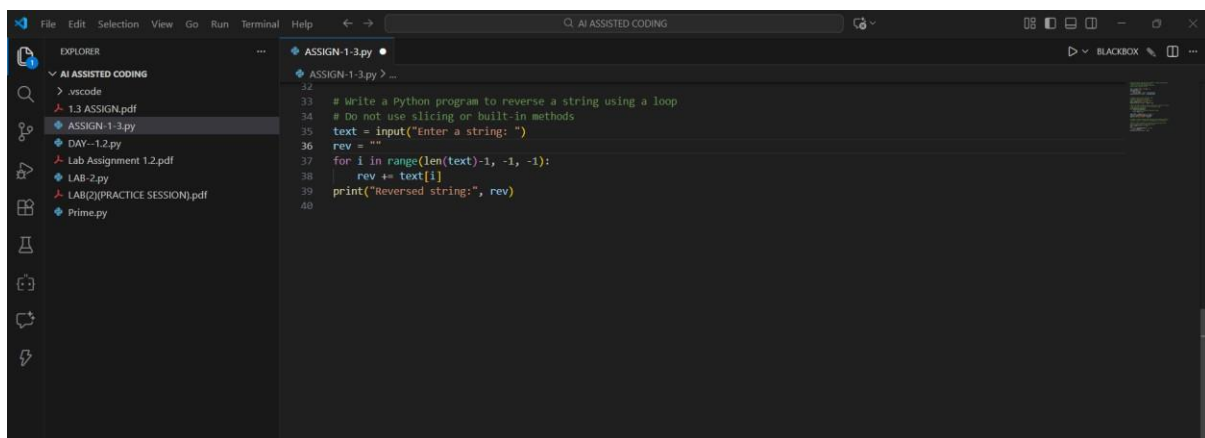
Task 5: Loop vs Built-in Reversal

Step 1: Loop-Based Prompt

Write a Python program to reverse a string using a loop

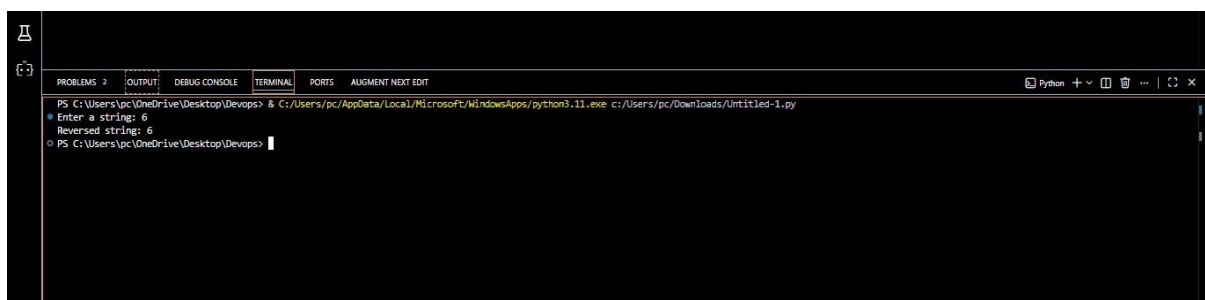
Do not use slicing or built-in methods

###CODE:

A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a file named 'ASSIGN-1-3.py' selected. The main editor window displays the code for 'ASSIGN-1-3.py'. The code is as follows:

```
33 # Write a Python program to reverse a string using a loop
34 # Do not use slicing or built-in methods
35 text = input("Enter a string: ")
36 rev = ""
37 for i in range(len(text)-1, -1, -1):
38     rev += text[i]
39 print("Reversed string:", rev)
```

OUTPUT:

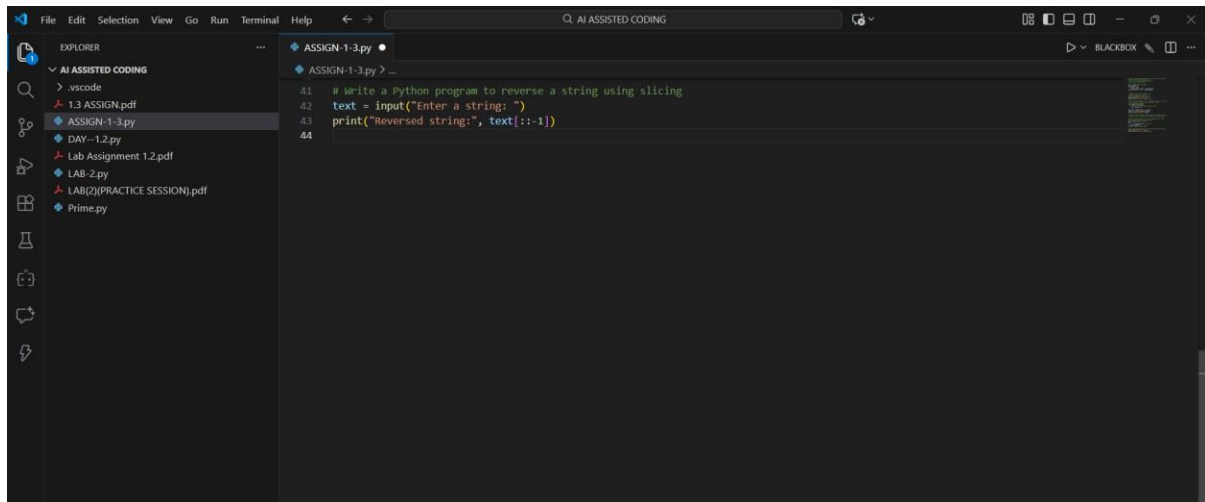
A screenshot of a terminal window. The terminal shows the execution of the Python program. The prompt 'PS C:\Users\pc\OneDrive\Desktop\Devops>' is followed by the command 'c:\Users\pc\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:\Users\pc\Downloads\Untitled-1.py'. The output shows 'Enter a string: 6' followed by 'Reversed string: 6'. The prompt 'PS C:\Users\pc\OneDrive\Desktop\Devops>' is shown again at the end.

```
PS C:\Users\pc\OneDrive\Desktop\Devops> & c:\Users\pc\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:\Users\pc\Downloads\Untitled-1.py
Enter a string: 6
Reversed string: 6
PS C:\Users\pc\OneDrive\Desktop\Devops>
```

Step 2: Built-in Prompt

Write a Python program to reverse a string using slicing

###CODE:



The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left lists files including '1.3 ASSIGN.pdf', 'ASSIGN-1.3.py', 'DAY-1.2.py', 'Lab Assignment 1.2.pdf', 'LAB-2.py', 'LAB(2)(PRACTICE SESSION).pdf', and 'Prime.py'. The main editor window is open to 'ASSIGN-1.3.py' and contains the following Python code:

```
41 # Write a Python program to reverse a string using slicing
42 text = input("Enter a string: ")
43 print("Reversed string:", text[::-1])
44
```

OUTPUT:



The screenshot shows the terminal window in VS Code. The command prompt is at 'PS C:\Users\pc\OneDrive\Desktop\Devops>'. The user has entered 'Enter a string: 24', and the program has output 'Reversed string: 42'.

```
PS C:\Users\pc\OneDrive\Desktop\Devops> & C:/Users/pc/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/pc/Downloads/Untitled-1.py
Enter a string: 24
Reversed string: 42
PS C:\Users\pc\OneDrive\Desktop\Devops>
```

Explanation: Two different string reversal approaches were generated using Copilot: loop-based and built-in slicing. Both methods have the same time complexity, but the built-in approach is more concise and readable.